

Optical Simulation of Scintillator Bars EJ-204 vs EJ-230 for the SHiP Timing Detector

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Scintillator bar (GEANT4):

- Active length: 1400 mm (± 700 mm along x)
- Coating: reflective Mylar
- $v_{\text{eff}} \approx 277$ mm/ns (nominal axial average)

→ ToF onset measures $v_{\text{eff}}^{\text{onset}} \approx 292$ mm/ns (+5%): set by the most direct rays, not the mean path (see QA-1c)

- Two materials compared in parallel:

EJ-204 vs **EJ-230**

End SiPMs (primary timing):

- 16 SiPMs total (IDs 0–15)
- END_LEFT: IDs 0–7 at $x = -700$ mm
- END_RIGHT: IDs 8–15 at $x = +700$ mm
- Timing estimator: **SUM4** (fixed id//4 clusters, earliest-of-two, leading-edge > 4 PE)

Notation used below: **END** = END_LEFT $\cup$ END_RIGHT (all IDs 0–15). “Left/right column” refers only to slide layout; physical sides are always written END_LEFT or END_RIGHT.

Top SiPMs (veto / tracking):

- 70 SiPMs, single row at $Y = +30.25$ mm
- TOP_LEFT: IDs 16–50 / TOP_RIGHT: 51–85
- Pitch: 20 mm in X ; central gap 24 mm

GEANT4 statistics:

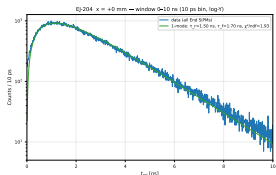
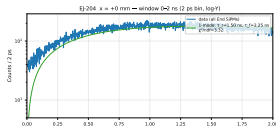
Dataset	Events	Positions
EJ-204 End	2000/pos	31
EJ-230 End	2000/pos	31
EJ-204 EndTop	2000/pos	31

Positions: $x = 0, \pm 50, \dots, \pm 690$ mm

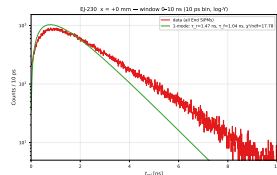
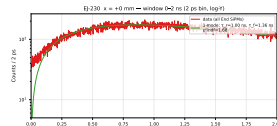
F1: Relative arrival time t_{rel} — central position $x = 0$

EJ-204 (left column) vs EJ-230 (right column) — each material pools both physical END faces

Here **END** = **END_LEFT** $\cup$ **END_RIGHT** (SiPM IDs 0–15). “Left/right” above describes the page columns, not the detector ends.



EJ-204 x=+0 mm, $t_{\text{rel}} = t_{\text{TOF}}$ — 100% lead percent



EJ-230 x=+0 mm, $t_{\text{rel}} = t_{\text{TOF}}$ — 100% lead percent

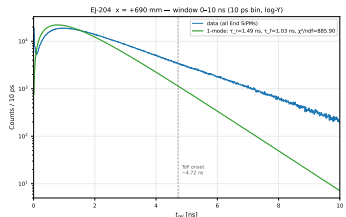
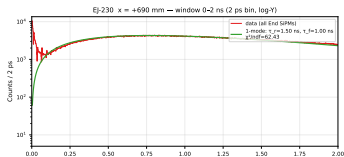
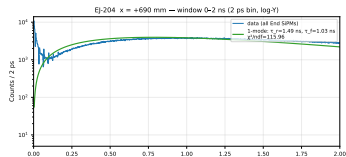
For event e : $t_{0,\text{END}}^{(e)} = \min_h: 1D(h)=0\dots15 \ t_h$; every hit on either END face enters as $t_{\text{rel}} = t_h - t_{0,\text{END}}^{(e)}$.

At $x = 0$, either physical end can provide the first hit. EJ-204: $\tau_r = 1.50$ ns, $\tau_f = 1.70$ ns ($\chi^2/\text{ndf} = 1.93$). EJ-230: $\tau_r = 1.47$ ns, $\tau_f = 1.04$ ns ($\chi^2/\text{ndf} = 17.8$).

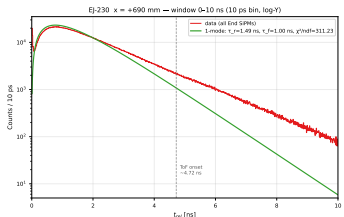
These fitted τ describe an arrival profile (emission $\otimes$ propagation $\otimes$ first-hit referencing), not datasheet constants. EJ-230's central tail cannot be a left-right ToF effect.

F1: Relative arrival time t_{rel} — edge position $x = +690$ mm

Both END faces are pooled: END_RIGHT is near, END_LEFT is far



EJ-204: $x = +690$ mm, $t_{\text{off}} = t_{\text{geom}} = \text{NHE}(t_{\text{off}})$ per event.
 Tail 3-10 ns: ~90% scattered near-end photons + ~2% far-end ToF IQA-1c onset=4.72 ns.

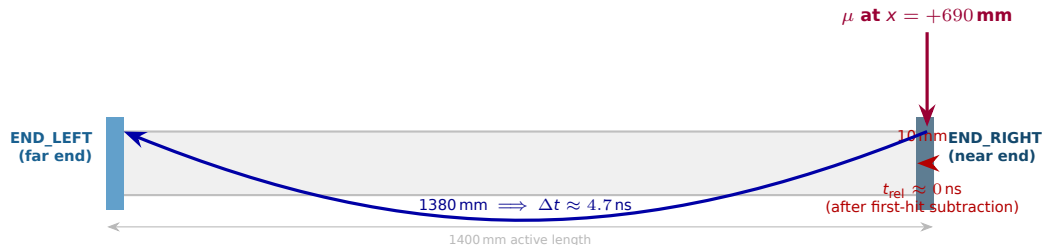


EJ-230: $x = +690$ mm, $t_{\text{off}} = t_{\text{geom}} = \text{NHE}(t_{\text{off}})$ per event.
 Tail 3-10 ns: ~90% scattered near-end photons + ~2% far-end ToF IQA-1c onset=4.72 ns.

The early population is dominated by the nearby END_RIGHT; the far END_LEFT starts contributing at $t_{\text{rel}} \simeq 4.7$ ns. Both populations are superposed in F1.

A 1-mode fit therefore fails ($\chi^2/\text{ndf} = 886$ for EJ-204 and 721 for EJ-230). The late onset is geometric ToF, not evidence by itself for slow scintillation.

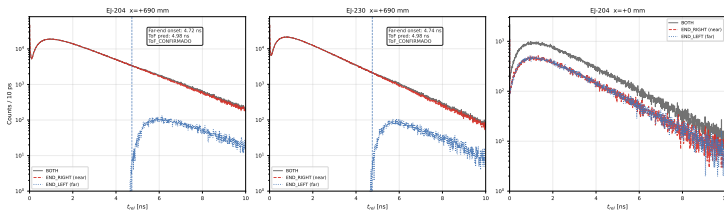
Why a second component appears at the edge — geometric time-of-flight



The late bump is the *same* photons reaching the *far end* — not slow scintillation.

QA-1c: Second F1 component confirmed as geometric ToF

QA-1c — t_{rel} per end | Grey=BOTH Red=END_RIGHT (near) Blue=END_LEFT (far)
Far end appears entirely at $t_{\text{rel}}=4$ ns (ToF confirmed, onset 4.72 ns vs 4.98 ns predicted)



BOTH is the union of photon hits from END_LEFT and END_RIGHT (grey = red + blue bin by bin), not a coincidence time. The labels near/far apply to $x = +690$ mm; at $x = 0$ neither side is geometrically near or far.

ToF prediction ($x = +690$ mm):

$$\Delta t_{\text{pred}} = \frac{1380 \text{ mm}}{277 \text{ mm/ns}} \approx 4.98 \text{ ns}$$

Measured onset (far end):

- EJ-204: 4.72 ns (−5.3%)
- EJ-230: 4.74 ns (−4.9%)

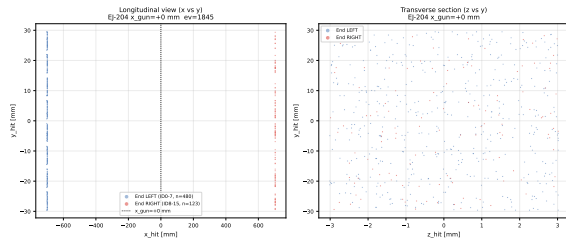
Verdict:

ToF_CONFIRMED

The −5% offset reflects $v_{\text{eff}}^{\text{onset}} \approx 292 \text{ mm/ns}$ — the fastest (most direct) optical rays arrive first, pulling the onset earlier than the mean-path prediction.

F3: Spatial distribution of photon impacts on SiPMs

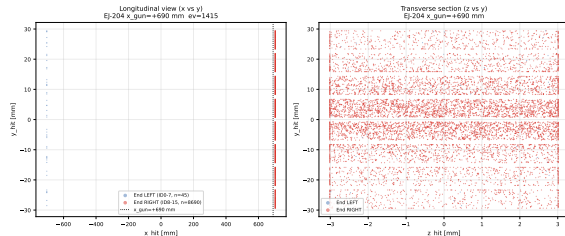
Single representative event — EJ-204 (EJ-230 qualitatively identical)



F3 — Photon hits on SiPMs (NOT volume tracks), event 1845 (603 hits). EJ-204 event display reconstructed from hits (not re-simulation).

$x_{\text{gun}} = 0$ mm — both ends populated; this single event is not count-symmetric

Note: Points are *photon impacts on the SiPM window* (NOT trajectories inside the scintillator volume). This is an endpoint map, not a ray-tracing view. Reconstructed from $x_{\text{mm}}/y_{\text{mm}}/z_{\text{mm}}$ ROOT branches — no re-simulation.

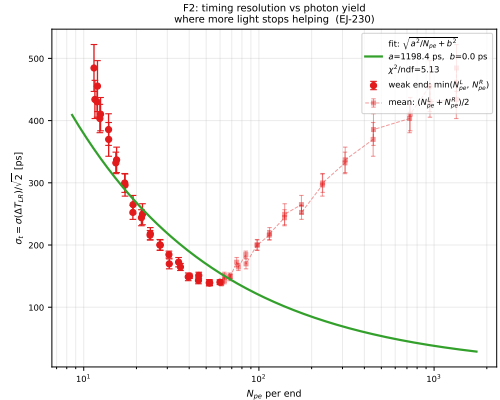
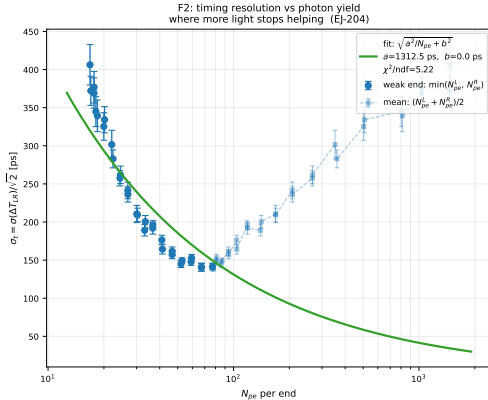


F3 — Photon hits on SiPMs (NOT volume tracks), event 1415 (8735 hits). EJ-204 event display reconstructed from hits (not re-simulation).

$x_{\text{gun}} = +690$ mm — collapse toward END_RIGHT

F2: σ_t vs N_{pe} — both END faces, analyzed separately

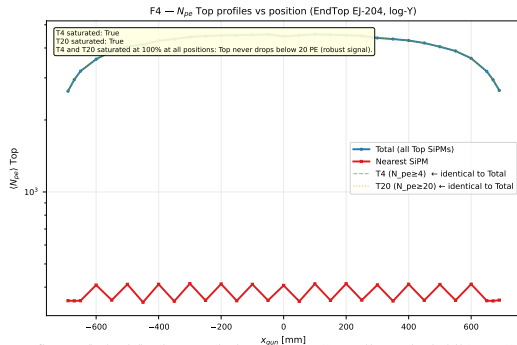
N_L : END_LEFT (IDs 0-7), N_R : END_RIGHT (IDs 8-15); fit variable $N_{\text{weak}} = \min(N_L, N_R)$



One position, two horizontal coordinates: dark markers use $N_{\text{weak}} = \min(N_L, N_R)$ and are fitted; faint markers use $N_{\text{mean}} = (N_L + N_R)/2$ with the same σ_t and are not fitted. For $x > 0$, weak/far = END_LEFT; for $x < 0$, weak/far = END_RIGHT; at $x = 0$ they are equivalent. Thus a huge near-end signal can inflate N_{mean} while the photon-starved far end still limits timing.

Fit: $\sigma_t = \sqrt{a^2/N_{\text{weak}} + b^2}$. Both fits place $b = 0$ over $N_{\text{weak}} \in [11, 77]$, but $\chi^2/\text{ndf} \simeq 5$ means the $1/\sqrt{N_{pe}}$ description is approximate, not exact “pure Poisson”.

F4: Top SiPM light yield vs position — not a timing quantity (EndTop EJ-204)

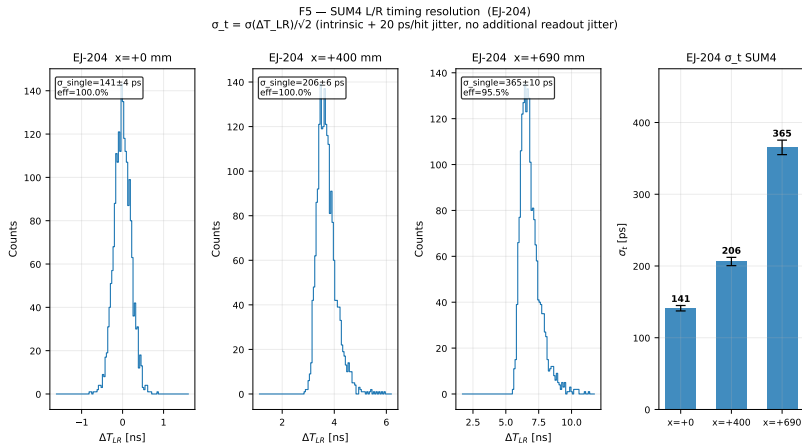


EJ-204 (EndTop) Top N_{pe} profiles vs x_{gun} (log-Y). Total: all Top SiPMs. Nearest: SiPM closest to the muon. T4/T20: events with $N_{pe_total} \geq$ threshold. OPEN: T4/T20 definition pending Gerardo.

- **TOP SiPMs** (IDs 16–85): Y-axis is light yield, not timing. “Total” sums all TOP channels; “Nearest” is the closest single channel and oscillates with the 20-mm pitch.
- All events pass the implemented T4/T20 selections, so those curves lie on Total: **100% cut efficiency**, not electronic sensor saturation.
- **Open — Gerardo:** confirm whether T4/T20 are total- N_{pe} cuts or a SUM4 trigger efficiency.

F5: SUM4 L/R timing resolution — EJ-204

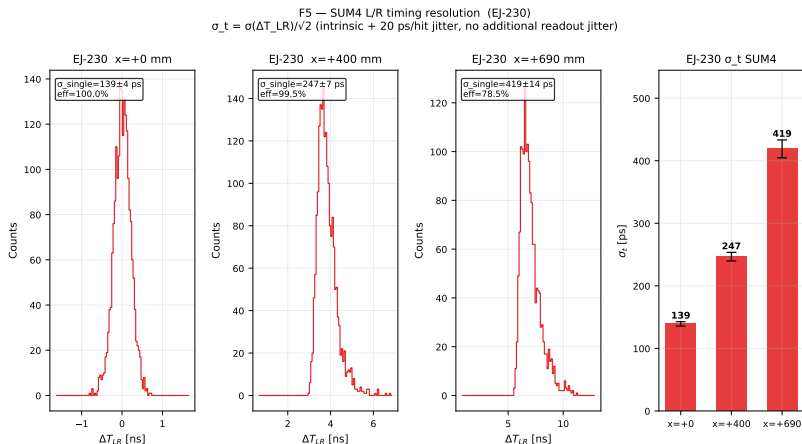
T_L : END_LEFT, T_R : END_RIGHT, $\Delta T_{LR} = T_L - T_R$; width gives timing, peak position gives geometry



At $x > 0$, END_RIGHT is nearer, so $T_R < T_L$ and the peak moves to positive ΔT_{LR} . The reported $\sigma_t = \sigma(\Delta T_{LR})/\sqrt{2}$ is an equivalent single-end benchmark; near the edge the two ends are not equally precise. Stage 1 reports the intrinsic timing resolution from photon counting and optical propagation, as produced directly by the simulation; no electronics or SiPM jitter is added.

F5: SUM4 L/R timing resolution — EJ-230

T_L : END_LEFT, T_R : END_RIGHT, $\Delta T_{LR} = T_L - T_R$; stronger edge broadening and efficiency loss than EJ-204



The peak shift with x is the propagation-time asymmetry; the peak width and non-Gaussian tail quantify timing fluctuations. At $x = +690$ mm the valid-event efficiency falls to 78.5%, indicating that the far END_LEFT often fails the SUM4 requirement.

Timing resolution comparison: EJ-204 vs EJ-230

Material	$x = 0 \text{ mm}$ $\sigma_t \text{ [ps]}$	$x = +400 \text{ mm}$ $\sigma_t \text{ [ps]}$	$x = +690 \text{ mm}$ $\sigma_t \text{ [ps]}$
EJ-204	141 ± 4	206 ± 6	365 ± 10
EJ-230	139 ± 4	247 ± 7	419 ± 14

Key observations:

- Central resolution **equivalent**: $141 \approx 139 \text{ ps}$
- At the edge, EJ-230 is 14.8% larger than EJ-204 (419 vs 365 ps; EJ-204 is 12.9% lower if referenced to EJ-230)
- EJ-230 degrades faster with position (+201% vs +159% from $x = 0$ to $x = 690$)
- Fitted τ_f : arrival-time profile (emission $\otimes$ propagation)
not datasheet constants
(EJ-204: $\tau_r^{(0)} = 0.7 \text{ ns}$, $\tau_d^{(0)} = 1.8 \text{ ns}$)

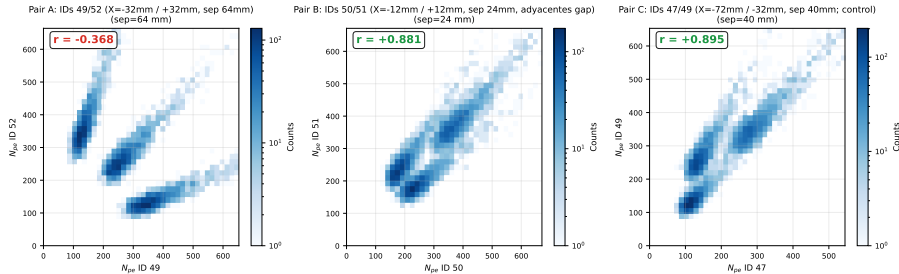
Caveats — Stage 1 (no real electronics):

- Intrinsic simulation only: no electronics or SiPM jitter added
- No time-walk correction (ToT)
- $\sigma/\sqrt{2}$ is an equivalent metric; edge ends are asymmetric
- SUM4 (code-verified): fixed id // 4 clusters per end, earliest-of-two, leading-edge $\geq 4 \text{ PE}$. Open: hardware fixed clusters vs moving-window.
- Recovering emission constants requires deconvolution — Stage 2

F6: Top SiPM redundancy — N_{pe} correlation between neighbors

F6 — Top SiPM redundancy: N_{pe} correlation between neighbouring pairs

Pair A (49/52, symmetric): $r = -0.37$ complementary | Pair B (50/51): $r = +0.88$ | Pair C (47/49, same side): $r = +0.89$ redundant



Pair A (IDs 49/52)

sep = 64 mm, opposite sides
 $r = -0.37$ — complementary
(light splits between sides)

Pair B (IDs 50/51)

sep = 24 mm, across gap
 $r = +0.88$ — **strongly correlated**
(same light source)

Pair C (IDs 47/49)

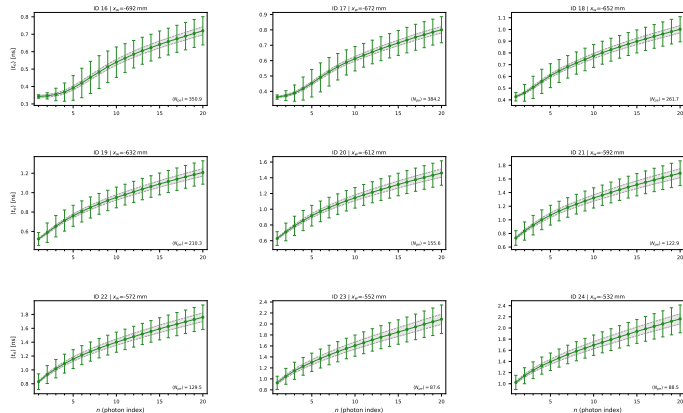
sep = 40 mm, same side
 $r = +0.90$ — **strongly correlated**
control
(local correlation expected)

Pearson r measures linear correlation only. High pooled r marks a candidate redundancy, not proof that a channel can be removed; fixed-position trigger/reconstruction tests are required.

F7: Order-statistic arrival time $\langle t_n \rangle$ per Top channel — $x = -690$ mm

Observed RMS vs idealized order-statistics floor $\sigma_{\text{stat}}(t_n) \simeq \sqrt{n} \tau_d / \langle N_{\text{pe}} \rangle$

F7 — Order-statistic arrival time (t_n) per Top channel | $x_{\text{gas}} = -690$ mm (EJ-204 EndTop)
Error bars = RMS (event-to-event). Dashed band: stat. floor $\sqrt{n} \tau_d / \langle N_{\text{pe}} \rangle$, $\tau_d = 1.8$ ns



How to read it:

- n : index of the n -th detected photon in one event/channel.
- $\langle t_n \rangle$: mean arrival time; it rises with n by construction.
- Error bars: event-to-event RMS, **not** uncertainty on the mean.
- Dashed band: idealized statistical floor using $\tau_d = 1.8$ ns.

RMS above the floor indicates an additional contribution compatible with optical-path dispersion. Widths must not be subtracted linearly; if independent,

$$\sigma_{\text{extra}} \simeq \sqrt{\sigma_{\text{obs}}^2 - \sigma_{\text{stat}}^2}.$$

IDs 16-24 span $x_w = -692 \rightarrow -532$ mm and are truncated by the left edge. The connection to F2 is qualitative: F7 uses TOP channels, whereas F2 uses END timing.

Main results:

- 1 **EJ-204** $\approx$ **EJ-230** at center (141 vs 139 ps); EJ-204 **better at edges** (365 vs 419 ps at $x = 690$)
- 2 F1 component at ~ 4.7 ns at $x = 690$ = **geometric ToF** from far end (QA-1c, $\delta = -5\%$). Not slow scintillation.
- 3 $v_{\text{eff}}^{\text{nominal}} = 277$ mm/ns; onset gives $v_{\text{eff}}^{\text{onset}} \approx 292$ mm/ns (+5%) — fastest optical paths set the onset
- 4 F2: timing follows a photon-statistics-dominated trend versus the weak end; no nonzero constant floor is resolved, although the simple fit has $\chi^2/\text{ndf} \sim 5$
- 5 F7: the observed t_n RMS exceeds an idealized statistical floor; the excess is evidence for an additional contribution compatible with optical-path dispersion (quadrature subtraction is required)
- 6 Top: all events pass the implemented 20-PE cut; gap-adjacent SiPMs are strongly correlated ($r \approx 0.88$), a candidate redundancy

Open — Gerardo confirmation:

- ☒ SUM4 (code-verified): fixed id//4 clusters, earliest-of-two, leading-edge ≥ 4 PE. Open: hardware fixed vs moving-window.
- ☐ T4/T20: N_{pe} cut or trigger efficiency
- ☒ Jitter: intrinsic simulation used as-is; no electronics or SiPM jitter term added.
- ☐ F3: SiPM impacts vs volume trajectories
- ☐ Pair A ($r = -0.37$): geometric or reflections
- ☐ EJ-230 tail at $x = 0$: slow mode or reflections?

Next steps:

- EJ-230 EndTop dataset (Top response)
- Stage 2: time-walk, readout jitter, real electronics
- Deconvolution: emission constants from arrival profiles